Occupancy Grids

Every cell has an occupancy probability

Log Odds $P(\text{Cell Occupied} \mid \text{history}, \text{sensor readings})$

$\rightarrow P(m \mid x_{1:t}, z_{1:t}) = P(\overset{x}{m} \mid \overset{y}{x_{1:t}}, \overset{z}{z_{1:t-1}}, z_t)$ Apply $P(x \mid y, z) = \dfrac{P(y \mid x, z) P(x \mid z)}{P(y \mid z)}$

$$= \frac{P(z_t \mid m, z_{1:t-1}, x_{1:t}) P(m \mid z_{1:t-1}, x_{1:t})}{P(z_t \mid z_{1:t-1}, x_{1:t})} = \frac{P(\text{reading} \mid \text{occupied}) \cdot P(\text{occu} \mid \text{history})}{P(\text{history} \mid \text{reading})}$$

Now apply markov property: Only current state matters, not history

- $P(z_t \mid m, z_{1:t-1}, x_{1:t}) = P(z_t \mid m, x_t)$
- $P(m \mid z_{1:t-1}, \underline{x_{1:t}}) = P(m \mid z_{1:t}, \underline{x_{1:t-1}})$

No sensor reading at t, so state is meaningless

$$= \frac{\overbrace{P(z_t \mid m, x_t)} P(m \mid z_{1:t-1}, x_{1:t-1})}{P(z_t \mid z_{1:t-1}, x_{1:t})}$$

Baye's Rule Again

- $P(z_t \mid m, x_t) = \dfrac{P(m \mid z_t, x_t) \cdot P(z_t \mid x_t)}{P(m \mid x_t)}$ ←

To get $P(m)$ we need state and sensor, state only is useless

$$= \frac{P(m \mid z_t, x_t) \cdot P(z_t \mid x_t)}{P(m)}$$

Indep Assumption

$$\overset{①}{=} \frac{P(m \mid z_t, x_t) P(z_t \mid x_t) P(m \mid z_{1:t-1}, x_{1:t-1})}{P(z_t \mid z_{1:t-1}, x_{1:t}) P(m)}$$

We are looking for a binary option, thus

② • $P(\neg m \mid x_{1:t}, z_{1:t}) = \dfrac{P(\neg m \mid z_t, x_t) P(z_t \mid x_t) P(\neg m \mid z_{1:t-1}, x_{1:t-1})}{P(z_t \mid z_{1:t-1}, x_{1:t}) P(\neg m)}$

Then Div ① by ②

$$\frac{P(m \mid x_{1:t}, z_{1:t})}{P(\neg m \mid x_{1:t}, z_{1:t})} = \frac{P(m \mid z_t, x_t)\, P(m \mid z_{1:t-1}, x_{1:t-1})\, P(\neg m)}{P(\neg m \mid z_t, x_t)\, P(\neg m \mid z_{1:t-1}, x_{1:t-1})\, P(m)}$$

$$= \frac{P(m \mid z_t, x_t) \quad P(m \mid z_{1:t-1}, x_{1:t-1}) \quad P(\neg m)}{\underbrace{(1-P(m \mid z_t, x_t))}_{\text{sensor update}}\ \underbrace{(1-P(m \mid z_{1:t-1}, x_{1:t-1}))}_{\text{history}}\ \underbrace{(1-P(\neg m))}_{\text{prior}}}$$

Great! Take log of all this.

$$l(m) = \log\left[\frac{P(m \mid z_t, x_t)}{1-P(m \mid z_t, x_t}\right] + \log\left[\frac{P(m \mid z_{1:t-1}, x_{1:t-1})}{1-P(m \mid \ldots}\right] + \log\left[\frac{1-P(m)}{P(m}\right]$$

Odds $\quad \text{Odds}(m) = \frac{P}{1-P} \rightarrow P = \frac{1}{1+\frac{1}{\text{odds}}}$

Log Odds $\quad l(m) = \log\left(\frac{P}{1-P}\right) \rightarrow P = 1 - \frac{1}{1+\exp l(m)}$

$$l(m) = \underbrace{l(m \mid z_{t:} x_t)}_{\text{sensor update}} + \underbrace{l(m_{t-1})}_{\text{history}} - \underbrace{l_0}_{\text{prior}}$$

$\therefore$ P(cell occu) = Prior log Odds + P Occu given Sensor − Prior

Bresenham Ray Casting